

1 - PURPOSE

The development of high dose sustained release tablets for water-soluble drugs can frequently require the use of large amounts of one or more matrix forming excipients resulting in overly large tablet sizes and/or onerous industrial processes inducing additional costs, such as coating.

Solid dispersion in lipid excipient enables water soluble active ingredients, having high melting point, to be completely embedded into an insoluble matrix which efficiently retards release.

This study describes the production of a solid dispersion containing Nicotinic acid (Niacin) and Compritol® 888 ATO (glyceryl behenate) by two different techniques: hot melt extrusion (HME) and a simple lab-scale Melt and Mix method.

2 - INTRODUCTION

Hot Melt Extrusion is the process of converting a raw material into a product of uniform shape and density by forcing it through a die under controlled conditions. The extruder consists of a feeder, a twin screw of various length/screw elements (barrel modules) and a die, where apart from the feeder all elements can be independently heated.

HME has been used for the development of various solid dosage forms aiming to increase the solubility of poorly soluble drugs, taste mask bitter substances and control drug release mainly by using thermoplastic polymers after the addition of plasticizers.

Lipid excipients are showing a new dimension to their utility in the form of making extrusion feasible at low temperature and without the addition of plasticizers. Described as "cold" extrusion, the continuous process enables active ingredients to be extruded with lipid excipients at processing temperatures below the lipid melting point (40° - 65°C depending on the excipient).

The lipid excipients can be used as release modifiers, with the advantage of enabling high drug loading; solubility enhancers and plasticizers for improved processing depending on the type of excipient used.

3 - EXPERIMENTAL METHODS

Nicotinic acid (Niacin; Vitamin B3) is a freely water soluble drug (27.2 mg/mL in 0.1N HCl at 25°C) which exhibits a very short half-life time of 20 to 45 minutes and that is used to reduce cholesterol blood levels and to decrease the risk of cardiovascular events. It is generally available in sustained release dosage forms with high doses from 500 mg to 1 g. The challenge for this drug is to get a reasonable tablet size that insures a prolonged drug release.

Compritol® 888 ATO (Glyceryl behenate - USP/NF) is a lipid excipient frequently used in sustained release applications. It is composed of glycerol behenate exhibiting a HLB of 1 and a melting point of 70°C. Due to its lipophilic nature and reduced wettability, drug delivery can be successfully extended over prolonged periods of time and might be highly effective even for freely water soluble drugs.

Binary mixtures of Compritol® 888 ATO and Niacin were extruded using the Leistritz Nano 16 twin-screw extruder at temperatures below the lipid melting point (<70°C) (Figure 1). The extrudates were macroscopically analyzed, prior to grinding and compression into tablet (Figure 2).



Figure 1: Twin screw extruder Nano 16 from Leistritz.

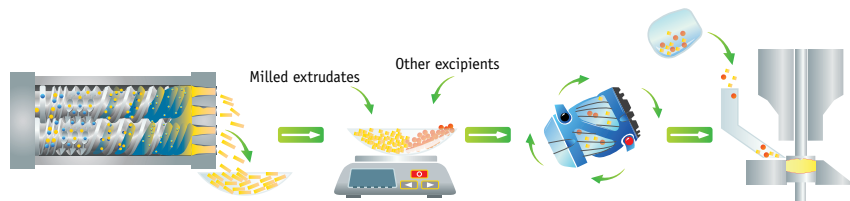


Figure 2: Hot melt extrusion followed by compression.

In parallel, a standard "lab-scale" solid dispersion called "Melt and Mix method", is made by dispersing Niacin in melted Compritol® 888 ATO. Once cooled, the mass was ground using an oscillating calibrator (Figure 3).

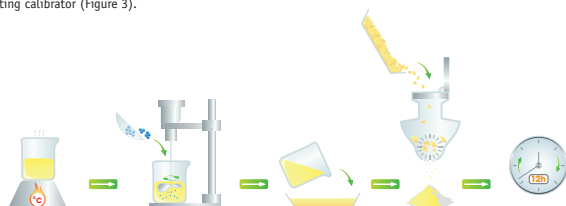


Figure 3: Melt and Mix method.

Granules are then compressed into tablets (Figure 4).

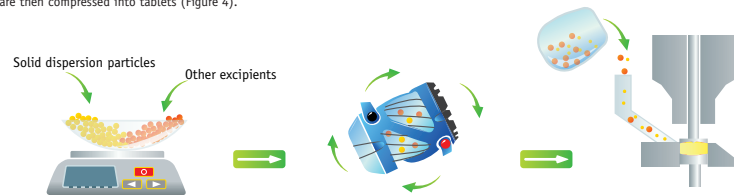


Figure 4: Compression of granules.

4 - RESULTS AND DISCUSSION

Compritol® 888 can be extruded without any additives and performs very well in melt extrusion, producing crystalline, brilliant and white strands at low processing temperature (60-65°C) (Figure 5a). The extrusion of a mixture of Compritol® 888 and Niacin (40:60) produces smooth, mat and white strands (Figure 5b).

Extrudates made by HME or granules made by Melt and Mix method were compared with a ratio Compritol® 888 ATO/Niacin = 60/40. Tablets were made without any additional excipient. Drug release from tablets containing extrudates made by HME was slower compared to tablets containing granules made by the Melt and Mix method (Figure 6). It is likely that mechanical mixing of the material by the twin-screw elements enables a better embedding of the drug in the melted Compritol®.

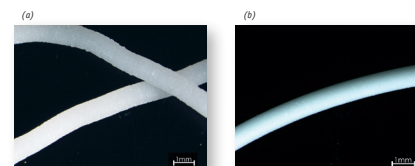


Figure 5: Lipid extrudates containing (a) Compritol® 888 - (b) Compritol®/Niacin 40:60.

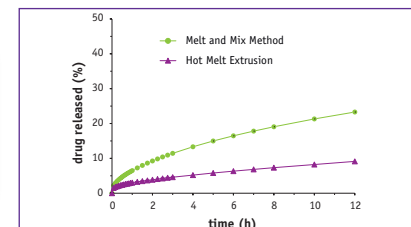


Figure 6: Tablets containing the solid dispersion granules prepared by the melt and mix method or HME and containing Compritol®/Niacin 60:40.

Although the Melt and Mix method enables the production of granules with good release retardant properties, high drug loading can be associated with increased viscosity preventing the production of a homogenous solid dispersion. This limitation is avoided using HME, since it generates high shear forces on the molten mixture with an excellent mass transfer behavior which is thus not affected by viscosity. Therefore, formulations with up to 60% drug load were processed by HME. For actives that require high drug dosing, such as Niacin, this is very interesting because the optimal ratio of drug versus matrix former is a key element to obtain a reasonable tablet size (Figure 7).

Figure 8 shows however, that increasing drug load and decreasing the Compritol® 888 results in faster drug diffusion.

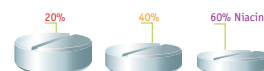


Figure 7: Schematic presentation of reduced tablet dimension with increasing drug load.

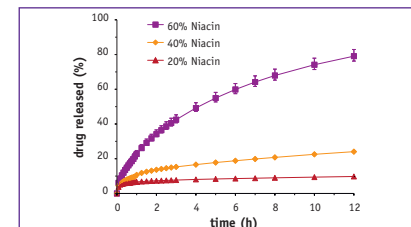


Figure 8: Effects of the drug load on release from Compritol®/Niacin extrudates.

5 - CONCLUSION

Continuous process twin-screw extrusion is an effective approach for the production of solid dispersions of Niacin in Compritol® for sustained release purposes. The approach is interesting for highly water soluble drugs and high drug loading formulations.